Applications of **simulation**-based burn probability modelling: a **review**

MA Parisien, DA Dawe, C Miller… - … of **wildland fire**, 2019 - CSIRO Publishing

… takes advantage of fire **spread** algorithms created for operational uses and the proliferation of available data representing **wildfire** patterns, fuels and weather. This **review** describes this …

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RY Zakari, K Kalinaki, ZK Lawal… - Artificial Intelligence and … - taylorfrancis.com

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SW Taylor, DG Woolford, CB Dean, DL Martell - 2013 - projecteuclid.org

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AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

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JL Coen, M Cameron, J Michalakes… - Journal of Applied …, 2013 - journals.ametsoc.org

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MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer

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P Johnston, G Milne, D Klemitz - **BUSHFIRE** CRC Project B, 2005 - naturalhazards.com.au

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N Khakzad - Reliability Engineering & System Safety, 2019 - Elsevier

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Z Zheng, W Huang, S Li, Y Zeng - Ecological Modelling, 2017 - Elsevier

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M Palk, K Knappmann, S Voss, R Jovanovic - IEEE Access, 2025 - ieeexplore.ieee.org

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